

Date	15 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing and indentation maintained throughout the project.	Yes	Improves readability and code consistency.
2	Proper File Structure	Files and folders are organized logically.	Yes	Templates, static files, dataset, models, and source code are separated into dedicated folders.
3	Meaningful Variable Names	Variables clearly describe their purpose.	Yes	Descriptive names such as prediction, features, and model are used.
4	Function / Method Names	Functions are descriptive and easy to understand.	Yes	Functions like predict() and model loading routines are clearly named.
5	Code Comments	Important sections contain comments explaining the logic.	Yes	Comments are provided for preprocessing, model training, and prediction flow.
6	Modular Design	Code is divided into reusable modules and functions.	Yes	Training script, Flask application, HTML templates, and CSS are maintained separately.
7	No Redundant Code	Duplicate or unnecessary code has been removed.	Yes	Code has been refactored to improve maintainability.
8	Error Handling	Exceptions and invalid inputs are handled gracefully.	Partial	Basic validation is implemented; additional exception handling can be added in future versions.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Flask Web Application	Python, Flask	User Interface & Prediction Workflow	High
2	Machine Learning Model	Scikit-learn	HDI Prediction	High
3	Data Preprocessing Module	Pandas, NumPy	Training & Data Cleaning	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
4	HTML Templates	HTML, Jinja2	Home Page & Result Page	Medium
5	CSS Styling	CSS	Responsive User Interface	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with a clear folder hierarchy.
Readability	5	Clean, descriptive, and easy-to-understand code.
Reusability	5	Modular components can be reused in similar ML web applications.
Documentation / Comments	4	Good code documentation with scope for more detailed API/function comments.
Overall Score	4.8 / 5	The project follows good coding practices and is maintainable, readable, and reusable.